

Carleson-Packed Product-BMO Lower Bounds for Bi-Commutators

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Status and source question

Status: candidate substantial partial result, likely valid; human review requested.

Hytönen records in Section 1.6(2), printed page 7, that the diagonal $L^p \rightarrow L^p$ characterization of the bi-commutator

$$[T_1, [M_b, T_2]]$$

remains open after a gap was found in the Ferguson–Lacey theorem and its extensions [1]. The exact source passage is reproduced below.

- (2) Iterated commutators of the form $[[b, T_1], T_2]$, where b is a function of two variable x_1, x_2 , and each T_i is a Calderón–Zygmund operator acting in the variable x_i , play an important role in the theory of singular integrals and function spaces on product domains. A characterisation of the $L^p \rightarrow L^p$ boundedness of these bi-commutators remains open, after the recent discovery of a gap [23] in the celebrated Ferguson–Lacey theorem [11] and its extensions; nevertheless, various mixed-norm $L^{p_1}(L^{p_2}) \rightarrow L^{q_1}(L^{q_2})$ bounds with $(p_1, p_2) \neq (q_1, q_2)$ have been recently characterised by Airta et al. [2], by extending the methods of the present paper.

For the Hilbert transforms on $\mathbb{R}^2$, the missing direction is the lower bound

$$\|b\|_{\text{BMO}_{\text{prod}}} \lesssim \|[H_1, [M_b, H_2]]\|_{L^2 \rightarrow L^2}.$$

Airta, Hytönen, Li, Martikainen, and Oikari explain the false Fourier-quadrant step and prove the surviving lower bound by rectangular BMO for general nondegenerate kernels [2]. This packet upgrades that bound to product BMO for a new support-packing class, and separately settles all pure tensor symbols. The unrestricted problem remains open.

Idea of the proof

Rectangular BMO says that after a first-coordinate interval I is fixed, the remaining coefficients indexed by J form a one-parameter Carleson sequence with constant at most $B^2|I|$. For an arbitrary open set Ω , let Ω_I collect the J for which $I \times J \subset \Omega$. The slice coefficient mass is at most $B^2|I||\Omega_I|$.

The only issue is summing this over I . If the active I ’s form a Carleson family, layer cake bounds the sum by the packing constant times $|\Omega|$. This produces exactly the open-set Carleson

inequality defining product BMO. It identifies arbitrary depth of incomparable rectangles as the obstruction left in the full problem.

For $b = u \otimes v$, the same layer-cake argument is weighted by the Haar coefficients of u , themselves a Carleson sequence. Simultaneously, the bi-commutator factors into the tensor product of the two one-parameter commutators, giving a dimension-free result without any scale restriction.

Definitions

Fix a tensor-product dyadic or Meyer wavelet basis $\{w_I \otimes w_J : I, J \in \mathcal{D}\}$, and put

$$b_{I,J} = \langle b, w_I \otimes w_J \rangle.$$

The coefficient forms of dyadic product and rectangular BMO are

$$\begin{aligned} \|b\|_{\text{BMO}_{\text{prod}}}^2 &= \sup_{0 < |\Omega| < \infty} \frac{1}{|\Omega|} \sum_{I \times J \subset \Omega} |b_{I,J}|^2, \\ \|b\|_{\text{BMO}_{\text{rect}}}^2 &= \sup_{I_0, J_0} \frac{1}{|I_0||J_0|} \sum_{I \subset I_0, J \subset J_0} |b_{I,J}|^2. \end{aligned}$$

For a standard Meyer system these norms are equivalent to the corresponding continuous Chang–Fefferman and rectangular norms, with constants depending only on the fixed system [4].

Define the first-coordinate active family

$$\mathcal{I}_b = \{I \in \mathcal{D} : b_{I,J} \neq 0 \text{ for some } J\}.$$

Its packing constant is

$$\Lambda_1(\mathcal{I}_b) = \sup_{0 < |E| < \infty} \frac{1}{|E|} \sum_{I \in \mathcal{I}_b, I \subset E} |I|,$$

where $E \subset \mathbb{R}$ ranges over open sets. Define Λ_2 symmetrically.

Packing theorem

Theorem 1 (Support-packing upgrade). *If $\Lambda_1(\mathcal{I}_b) < \infty$, then*

$$\|b\|_{\text{BMO}_{\text{prod}}} \leq \sqrt{\Lambda_1(\mathcal{I}_b)} \|b\|_{\text{BMO}_{\text{rect}}}.$$

The symmetric estimate holds with Λ_2 . Hence the better of the two finite packing constants may be used.

Proof. Write $B = \|b\|_{\text{BMO}_{\text{rect}}}$. For fixed I and every dyadic J_0 ,

$$\sum_{J \subset J_0} |b_{I,J}|^2 \leq \sum_{I' \subset I, J \subset J_0} |b_{I',J}|^2 \leq B^2 |I| |J_0|. \quad (1)$$

Thus $\{|b_{I,J}|^2\}_J$ is a one-parameter Carleson sequence with constant at most $B^2 |I|$.

Fix an open $\Omega \subset \mathbb{R}^2$ of finite measure, and let Ω_I be the union of all dyadic intervals J such that $I \times J \subset \Omega$. Decompose Ω_I into maximal dyadic intervals. Applying (1) on these disjoint intervals gives

$$\sum_{J: I \times J \subset \Omega} |b_{I,J}|^2 \leq B^2 |I| |\Omega_I|. \quad (2)$$

Put $g_I = |\Omega_I|$ and $E_t = \bigcup\{I \in \mathcal{I}_b : g_I > t\}$. Layer cake and the packing hypothesis give

$$\begin{aligned} \sum_{I \in \mathcal{I}_b} |I| g_I &= \int_0^\infty \sum_{I: g_I > t} |I| dt \\ &\leq \Lambda_1 \int_0^\infty |E_t| dt. \end{aligned} \quad (3)$$

If $x \in E_t$, then $x \in I$ for an interval with $|\Omega_I| > t$, and Ω_I is contained, up to endpoints, in the vertical section Ω_x . Therefore $E_t \subset \{x : |\Omega_x| > t\}$, and Fubini gives

$$\int_0^\infty |E_t| dt \leq \int_{\mathbb{R}} |\Omega_x| dx = |\Omega|. \quad (4)$$

Summing (2), then using (3)–(4), proves

$$\sum_{I \times J \subset \Omega} |b_{I,J}|^2 \leq B^2 \Lambda_1 |\Omega|.$$

Taking the supremum over Ω completes the proof. The other coordinate is identical. $\square$

Corollary 2 (Finite one-coordinate scale complexity). *If the active wavelet coefficients use at most M distinct scales in either coordinate, then*

$$\|b\|_{\text{BMO}_{\text{prod}}} \leq \sqrt{M} \|b\|_{\text{BMO}_{\text{rect}}}. \quad (5)$$

Consequently, for symmetrically nondegenerate Calderón–Zygmund operators in the two coordinates,

$$\|b\|_{\text{BMO}_{\text{prod}}} \lesssim \sqrt{M} \|[T_1, [M_b, T_2]]\|_{L^2 \rightarrow L^2}. \quad (6)$$

Together with the standard product-BMO upper bound, (6) is a full boundedness characterization on this finite-complexity class.

Proof. At each fixed scale, the dyadic intervals contained in an open set are disjoint, so their lengths sum to at most the measure of the set. Hence an M -scale active family has packing constant at most M . Apply Theorem 1; then use the rectangular lower bound of [2, Proposition 5.3], whose off-support constant is dominated by the operator norm. $\square$

Pure tensor theorem

Theorem 3 (Rank-one symbols). *For one-parameter functions u, v ,*

$$\|u \otimes v\|_{\text{BMO}_{\text{prod}}} \asymp \|u\|_{\text{BMO}} \|v\|_{\text{BMO}}. \quad (7)$$

Moreover,

$$[T_1, [M_{u \otimes v}, T_2]] = [T_1, M_u] \otimes [M_v, T_2]. \quad (8)$$

In particular, for T_1, T_2 having the usual one-parameter commutator characterization of BMO (including the Hilbert transforms), diagonal bi-commutator boundedness is equivalent to product BMO on all pure tensor symbols, with no scale restriction.

Proof. Let $a_I = \langle u, w_I \rangle$ and $c_J = \langle v, w_J \rangle$. Testing the product-BMO norm on $I_0 \times J_0$ and approximating the two one-parameter suprema proves the lower bound in (7).

For the upper bound, fix Ω and define Ω_I as above. The Carleson property of $\{|c_J|^2\}$ gives

$$\sum_{J:I \times J \subset \Omega} |a_I c_J|^2 \leq |a_I|^2 \|v\|_{\text{BMO}}^2 |\Omega_I|. \quad (9)$$

Using layer cake again, now with the Carleson weights $|a_I|^2$, yields

$$\begin{aligned} \sum_I |a_I|^2 |\Omega_I| &= \int_0^\infty \sum_{I:|\Omega_I|>t} |a_I|^2 dt \\ &\leq \|u\|_{\text{BMO}}^2 \int_0^\infty |\{x : |\Omega_x| > t\}| dt \\ &= \|u\|_{\text{BMO}}^2 |\Omega|. \end{aligned} \quad (10)$$

Equations (9)–(10) prove the upper bound in (7).

Identity (8) follows directly because T_1 and T_2 act in separate coordinates. Operator norms multiply on Hilbert tensor products. The one-parameter commutator theorem [5] and (7) finish the proof. $\square$

Verification, upgrade attempts, and limitations

The proof is analytic and uses no computation. Eight focused upgrade routes were checked before promotion: current-literature status, summing all Fourier quadrants, modulation/reflection, replacing bounded overlap by Carleson packing, exploiting small rectangular BMO in the old bootstrap, Rademacher randomization, smooth symbol-frequency projections, and finite tensor-rank factorization. The first four produced Theorem 1; the last route produced Theorem 3. The remaining routes fail to control arbitrarily deep incomparable rectangle families uniformly.

Holmes–Treil–Volberg prove a different dyadic partial theorem under an alternating-scale condition on the Haar support [3]. That result does not contain the Carleson-packed support class and does not repair the continuous Hilbert-transform theorem. Conversely, this packet does not contain their infinite alternating-scale class. The unrestricted diagonal problem remains open.

A bounded search through August 11, 2026 covered the run’s four lightweight indexes, the local sources of [1, 3], the current version of [2], and official-arXiv queries using the exact problem and the terms “product BMO”, “rectangular BMO”, “finite scales”, and “Carleson family”. No matching support-packing statement was found. This is not a priority claim.

Human review should focus on the equivalence between the fixed wavelet rectangular norm and the oscillation norm used in [2], the maximal dyadic decomposition of Ω_I , and the weighted layer-cake step in Theorem 3.

References

- [1] T. Hytönen, *The L^p -to- L^q boundedness of commutators with applications to the Jacobian operator*, arXiv:1804.11167, especially Section 1.6(2).
- [2] E. Airta, T. Hytönen, K. Li, H. Martikainen, and T. Oikari, *Off-diagonal estimates for bi-commutators*, Int. Math. Res. Not. IMRN 2022, no. 23, 18766–18832; arXiv:2005.03548.
- [3] I. Holmes, S. Treil, and A. Volberg, *Dyadic bi-parameter repeated commutator and dyadic product BMO*, arXiv:2101.00763.

- [4] S.-Y. A. Chang and R. Fefferman, *A continuous version of duality of H^1 with BMO on the bidisc*, Ann. of Math. (2) 112 (1980), 179–201.
- [5] R. Coifman, R. Rochberg, and G. Weiss, *Factorization theorems for Hardy spaces in several variables*, Ann. of Math. (2) 103 (1976), 611–635.